

Custom Neural Machine Translation Engine Development – Korean to English for Forum Content (Pann, Instiz, TheQoo)

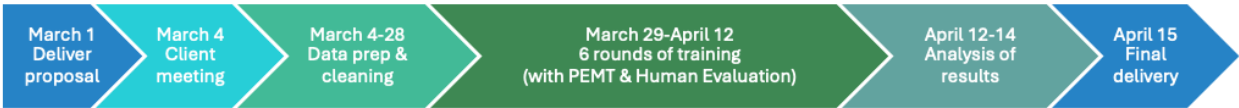


Project Objective

The purpose of this pilot project was to estimate the resources required to train a customized Neural Machine Translation (NMT) engine for KO-to-EN translations of forum-style K-pop content from Korean *netizens* (“internet citizens”) on Pann, Instiz, and TheQoo. A total of 6 rounds were conducted using the NMT engine Microsoft Azure. We did human post-editing and evaluation after each round.

Goals	Initial	Updated
Quality goals	• Meaning fully preserved • Informal tone/register accurate • Cultural & fandom terms clear • Light to moderate post editing required	• Meaning mostly preserved • Informal tone/register sometimes accurate • Cultural & fandom terms clear • Moderate to heavy post editing required
Timing goals	20%-30% faster than full HT	~20% faster than full HT
Pricing goals	20%-30% cost reduction vs. full HT via PEMT workflow	PEMT ~40% savings over HT

Updated timeline



Updated project costs

Task	Estimated	Actual	Rate	Subtotal
Data prep & cleaning	20 hours	50 hours	\$40	\$2,000
Rounds of training	10 rounds	6 rounds	\$40	\$240
Linguistic post-editing evaluation	7 hours	9 hours	\$40	\$360
Glossary collection	2 hours	1 hour	\$40	\$40
Subtotal				\$2,640

Recommended settings in a translation environment tool to assist in PEMT

1. Enabling a custom NMT model trained on domain-specific data, combined with enforced terminology through bilingual glossaries.
 - Helps ensure consistent and accurate translation of recurring forum-specific terms such as names, slang, and fandom references.
2. Proper sentence segmentation
 - Allows the system to better handle informal Korean structures and produce more coherent English output, thereby reducing the need for extensive rewriting.
3. Integrating automated quality assurance checks.
 - Examples include terminology compliance, formatting and tag verification, length mismatch detection, and grammar/spell checks.
 - Helps identify common machine translation errors early in the workflow.

When combined with translation memory integration and, where available, quality estimation for segment-level filtering, these settings collectively reduce post-editing effort, improve consistency, and support measurable gains in productivity and cost efficiency within the PEMT process.

Suggestions for further training of NMT

For future improvements, there are a few simpler things that could be done to make the project better. One important step would be to collect more examples of Korean forum slang with clear context, since many slang words change meaning depending on how they are used. It might also help to organize the data better or include short explanations of what each slang term means. Another idea is to keep updating the dataset over time, because new slang appears very quickly online. Finally, combining the model's output with some manual checking could improve accuracy, especially when the translations are unclear or confusing. These changes could make the system more reliable without making it too complicated.